Redundancy and Geography in Supply-Network Cascades:
Extending the Elliott–Golub–Leduc Model to k-of-n Sourcing and Regional Shocks

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Abstract

Elliott, Golub, and Leduc [1] (EGL) show that supply networks can collapse suddenly, Carvalho, Nirei, Saito, and Tahbaz-Salehi [2] (CNST) measure how the 2011 Japanese earthquake spread through real supply links, and Dorogovtsev, Goltsev, and Mendes [3] (DGM) explain why k-cores of random graphs vanish abruptly. We extend EGL's model in two ways: a firm needs at least k working suppliers for each input, and a regional disaster shuts down every firm in a region at once. We show that collapse becomes sudden as soon as a firm needs two of anything, that with one input it does not matter for the breaking point whether suppliers are spread out or co-located, and that spreading suppliers out never hurts when one supplier is enough. When firms need two suppliers, however, splitting them into two regional pairs can make the network break at a smaller shock than either extreme. Simulations of networks with 420,000 firms agree closely with these predictions.

1 Introduction

Supply networks are a natural setting for the cascades studied in this course: a firm missing one essential input cannot produce, and its customers then face the same problem. Firms protect themselves in two ways: they buy each input from several suppliers (*redundancy*), and they place those suppliers in different regions (*geography*). We ask how these two protections move the point at which a supply network collapses.

We read three papers, none linked from the course page. EGL is our starting point: it models supply networks as random trees and finds a "precipice" in output. CNST is the leading empirical study of supply-chain contagion, and the shock it studies is regional. DGM analyzes k-cores, which collapse abruptly for the same mathematical reason as our model. Sections 2–5 discuss and connect the papers, Section 6 proposes and analyzes our model, Section 7 tests it in simulation, and Section 8 outlines next steps.

2 Paper 1: Elliott, Golub, and Leduc (2022), "Supply Network Formation and Fragility"

Technical content. Firms sit in layers of a deep random supply tree. A firm needs m essential inputs and has n potential suppliers for each in the layer below; each relationship works independently with probability x (*relationship strength*). A firm is functional if every input has at least one working relationship with a functional supplier. If each supplier is functional with probability r, a firm is functional with probability

$$R_x(r) = \left(1 - (1 - xr)^n\right)^m, \tag{1}$$

and in a deep network the *reliability* (the probability that a firm works) is the largest fixed point of R_x. For $m \geq 2$, R_x is convex then concave, so as x falls, the largest crossing with the diagonal disappears at a tangency, and reliability *jumps* to zero at a critical strength x_{crit} (their Proposition 1). For $m = 2$ and $n = 4$ we compute $x_{\text{crit}} = 0.518$, where reliability drops from 0.64 to 0. For $m = 1$ the drop is continuous and happens at $x = 1/n$. EGL then let firms pay to raise their own x. Firms ignore how their reliability helps their customers, so for $m \geq 3$ a whole range of cost parameters puts the equilibrium just above x_{crit} (their Theorem 1), where a tiny drop in baseline strength destroys production.

Relevance to the course. The failure rule is a threshold rule like the adoption rule of Granovetter [6] and Easley–Kleinberg [5, §19.2], run in reverse: a firm drops out when, for some input, none of its suppliers remains. With $m = 1$ the recursion is the survival probability of a branching process [5, §21.8]. With $m \geq 2$ it resembles a complex contagion [4]. The new element, in the spirit of [8], is that firms' own choices set how close the network sits to its threshold.

Weaknesses and improvements. (i) Relationship failures are independent, and the only aggregate shock is a uniform drop in x. Real disasters are regional, so co-located suppliers fail together. (ii) One working supplier always suffices, which ignores capacity limits. (iii) The tree has no hubs or cycles, the equilibrium theorem needs $m \geq 3$, and x_{crit} cannot be observed in data. Our model addresses (i) and (ii), and we test supplier hubs in simulation.

3 Paper 2: Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021), "Supply Chain Disruptions"

Technical content. The 2011 earthquake and tsunami hit four prefectures holding 4.6% of Japan's output, whose own losses directly account for only about 0.1 percentage points (pp) of national growth. Using supplier–customer links among 750,237 firms (about 5 suppliers each, heavy-tailed degrees) and a difference-in-differences design, CNST find that sales growth fell by 3.8pp for firms with suppliers in the disaster area and 3.1pp for firms with customers there, shrinking with network distance and robust to a placebo network that randomly rewires links. Yet their calibrated model attributes 0.47pp of lost GDP growth to the disaster, versus 0.21pp had links out of the region been cut.

Relevance, weaknesses, and improvements. This is a clean causal measurement of contagion on a real network, and the effect fading with each hop is consistent with smooth, damped propagation. Their model spreads a shock through a damped sum over all supply paths, with weights that shrink with path length, much like the walk sums behind PageRank [5, §14.6]. Three weaknesses limit what it can say about collapse. First, production in the model substitutes smoothly between inputs and firms never shut down, so output falls smoothly with the shock and cannot show a threshold. Second, exposure is measured as network distance to the disaster area, which treats a firm with 1 of 10 suppliers there the same as one with 10 of 10. Third, the sample drops firms that exited, and exit is the outcome most telling of collapse. Section 8 proposes a test that addresses all three.

4 Paper 3: Dorogovtsev, Goltsev, and Mendes (2006), "k-Core Organization of Complex Networks"

Technical content. The k-core of a graph is what remains after repeatedly deleting every vertex with fewer than k neighbors. DGM compute its size in random graphs with a given degree distribution (the configuration model) after a random fraction $Q = 1 - p$ of vertices is removed. Because these graphs look locally like trees, this reduces to a fixed-point equation for the probability u that an edge leads to a vertex that can stay in the core; for an Erdős–Rényi (ER) graph with mean degree z, $u = p\Pr[\text{Poisson}(zu) \geq k-1]$. For $k = 2$ the core shrinks continuously, like the giant component; for $k \geq 3$ it vanishes with a *jump*. Just before collapse the core is held together by vertices with exactly k core neighbors, so one more deletion sets off a large avalanche of pruning. Solving their equations for $z = 10$, we find the 7-core vanishes once 1.25% of vertices are removed, while the giant component survives until 90% are gone.

Relevance, weaknesses, and improvements. The analysis uses the configuration model and branching arguments from the Random Graphs part of the course, and pruning a core is itself a threshold cascade. But it is mean-field (ignoring short cycles and degree correlations), damage is random and independent, every vertex has the same threshold, and links are undirected. A supply-network version needs directed links, firm-specific thresholds, and correlated damage; our model adds the first and last.

5 Synthesis and Research Questions

EGL and DGM both find that a network collapses suddenly, rather than gradually, when each node needs more than one thing to keep working: EGL's firms need several inputs, and DGM's core vertices need several neighbors. CNST, by contrast, see smooth propagation after a large regional shock. The two views fit together, because CNST's model has no way to produce a jump, they only measure firms that survived, and in EGL's model a network far above its tipping point also absorbs shocks smoothly.

Two features of real supply chains are missing from all three papers. First, a firm often needs more than one working supplier for an input, because of capacity limits or dual-sourcing policies. Second, suppliers in the same region fail together. EGL and DGM assume independent failures, and CNST's model has no backup suppliers. We therefore ask: **Q1:** If a firm needs $k \geq 2$ working suppliers for an input, does collapse become sudden even when firms have only one input? **Q2:** Is it safer to spread a firm's suppliers across regions or to keep them together, and does the answer depend on k? **Q3:** When a region is hit, does the number of firms it holds or the amount of supply it provides

matter more for collapse? Sections 6 and 7 are our initial steps on these questions, and Section 8 describes how we plan to continue in the final project.

6 Our Model: k-of-n Sourcing with Regional Shocks

Setup. We keep EGL's layered tree. Each firm needs m inputs, has n potential suppliers for each in the layer below, and each relationship works with probability x. We make two changes.

- **k-of-n sourcing.** An input is covered only if at least k of its n suppliers are *usable*, meaning the supplier works and so does the relationship. EGL is the case $k = 1$.
- **Regional shocks.** Firms are spread over many regions. A disaster hits each region independently with probability ϕ and shuts down every firm in it. A firm's n suppliers for an input form c clusters of n/c; suppliers in a cluster share a region, and different clusters are (almost always) in different regions. With $n = 4$ we compare *spread* suppliers ($c = 4$), two *pairs* ($c = 2$), and *co-located* suppliers ($c = 1$).

Let r be the probability that a firm works given that its own region was not hit, and let $s = 1 - \phi$. A supplier whose region survived is usable with probability $p = xr$. Write $S_k(N, p)$ for the chance that at least k of N such suppliers are usable (a binomial tail). Then

$$r = F(r) := \big(\Pr[\text{an input is covered}]\big)^m, \tag{2}$$

where the coverage probability depends on the geography. With spread suppliers, each supplier survives independently, so it is $S_k(n, sxr)$. With co-located suppliers, all n survive together with probability s, so it is $s\,S_k(n, xr)$. With two pairs, both pairs survive with probability s^2 and exactly one does with probability $2s\phi$, so it is $s^2 S_k(4, xr) + 2s\phi\, S_k(2, xr)$. As in EGL, the reliability of a deep network is the largest solution r^* of (2), and the share of all firms that work is $\rho = s\,r^*$. The *tolerable shock* ϕ^* is the largest ϕ at which $r^* > 0$. For $m = 1$ and spread suppliers, (2) has the same form as DGM's equation for the $(k+1)$-core (a core vertex needs k neighbors besides the one it was reached from, as our firm needs k suppliers), so we expect sudden collapse once $k \geq 2$.

Claim 1 (Abrupt collapse)**.** *If $km \geq 2$, then as x falls or ϕ rises, reliability r^* jumps from a positive value straight to 0. If $k = m = 1$, it shrinks to 0 gradually.*

Proof. The chance that at least k of the suppliers are usable is at most $\binom{n}{k} p^k$ (add up the chances over all groups of k suppliers). So $F(r) \leq K r^{km} \leq K r^2$ with $K = \binom{n}{k}^m$. For $0 < r < 1/K$ this gives $F(r) < r$, so (2) has no solution in $(0, 1/K)$: reliability is either 0 or at least $1/K$. Since r^* moves monotonically with x and ϕ, it must jump. If $k = m = 1$, then for small r we have $F(r) \approx nsx\,r$, which is a branching process with mean nsx; its survival probability grows from 0 continuously once $nsx > 1$ [5, §21.8]. □

For $m \geq 2$ this is EGL's precipice. The new case is $m = 1$ and $k \geq 2$, where needing two suppliers of a single input is already enough for a cliff. For example, with $m = 1$, $n = 4$, $k = 2$, reliability drops from 0.745 to 0 at $x = 0.725$ (Figure 1a).

Claim 2 (Spread versus co-located suppliers)**.** *(a) With spread suppliers, a shock ϕ has exactly the effect of weakening every relationship from x to $(1-\phi)x$, so $\phi^*_{\text{spread}} = 1 - x_{\text{crit}}/x$, where x_{crit} is the no-shock critical strength for this k. (b) With one input ($m = 1$), spread and co-located suppliers have the same tolerable shock. With m inputs, $\phi^*_{\text{co}} = 1 - (1 - \phi^*_{\text{spread}})^{1/m}$, which is smaller.*

Proof. (a) With spread suppliers, $F(r) = [S_k(n, sxr)]^m$, which is EGL's recursion with strength sx; it keeps working as long as $sx \geq x_{\text{crit}}$. (b) Let $G(u) = [S_k(n, xu)]^m$. Spread suppliers give $r = G(sr)$; multiplying by s and writing $u = sr$ gives $u = sG(u)$. Co-located suppliers give $r = s^m G(r)$. So both cases lead to the same equation $u = aG(u)$, and only the constant in front differs: $a = s$ for spread and $a = s^m$ for co-located suppliers. Raising a only raises the right-hand side, so if a positive solution exists for some a, one exists for every larger a. This gives a single cutoff a^* (depending only on G). Spread networks therefore collapse once s falls below a^*, and co-located networks once s^m does. □

With several inputs, co-location gives each input its own chance of being wiped out. For $m = 2$, $n = 4$, $x = 0.9$, co-location lowers ϕ^* from 0.425 to 0.242 when $k = 1$, and from 0.077 to 0.039 when $k = 2$. Requiring $k = 2$ instead of $k = 1$ shrinks the tolerable shock more than fivefold.

Claim 3 (More clusters when $k = 1$)**.** *If $k = 1$, splitting each cluster into smaller ones (going from c to c' clusters, where c divides c') never lowers r^* or ϕ^*.*

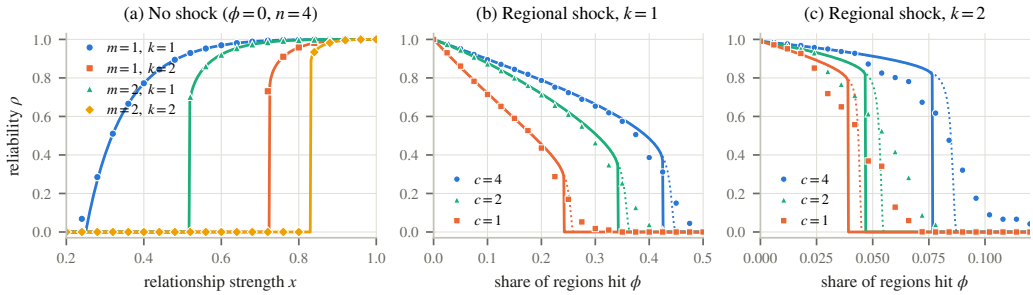


Figure 1: Simulation (markers) against theory (solid: infinite depth; dotted: 20 layers). (a) No shock, $n=4$: reliability against relationship strength; $k=2$ creates a jump even with one input (Claim 1). (b, c) Reliability against the share of regions hit for $m=2$, $n=4$, $x=0.9$, with $k=1$ and $k=2$, for each number of clusters c.

Proof (for splitting each cluster in two; other splits work the same way). With $k=1$, an input fails only if no surviving supplier is usable. Let $q=1-xr$. A cluster of b suppliers provides no usable supplier with probability $\phi+sq^b$: either it is hit, or it survives and all b fail. So $F(r)=[1-(\phi+sq^b)^c]^m$. Splitting each cluster into two halves replaces $\phi+sq^b$ by $(\phi+sq^{b/2})^2$, and since $y\mapsto y^2$ is convex, Jensen's inequality gives $(\phi\cdot 1+st)^2\le\phi\cdot 1^2+st^2$ with $t=q^{b/2}$. So each input fails less often, F goes up, and so does its largest fixed point. □

Example 1 (Pairs can be worst when $k=2$)**.** *Claim 3 does not extend to $k\ge 2$. With $n=4$, $k=2$, $m=1$, $x=0.9$, solving* (2) *numerically gives $\phi^*=0.195$ for co-located suppliers,* 0.171 *for two pairs, and* 0.195 *for spread suppliers.*

The two extremes tie by Claim 2(b). To see why pairs do worse, subtract the coverage formulas above: an input with two pairs is covered more often than a co-located one by $s\phi\,[2p^2-S_2(4,p)]$, where $p=xr$. A paired firm often loses exactly one pair (probability $2s\phi$) and must then use both remaining suppliers (probability p^2), while a co-located firm is either untouched or loses all four suppliers (probability ϕ). When relationships are strong, the bracket is positive and pairs are better. Near the tipping point p is only about 0.6, below the value $2/3$ where the bracket turns negative, and pairs are worse. These barely sourced firms are like DGM's vertices with exactly k core neighbors. In a grid search over even $n\le 8$, $m\le 3$, and six values of x, such reversals appeared in 72 of the 288 settings with $k\ge 2$ and in none with $k=1$. For Q2, then, splitting suppliers into more regions never hurts when one supplier is enough, but when firms need two, splitting them into two pairs can give the smallest tolerable shock.

7 Simulations

Setup. The claims are about infinitely deep trees, so we check them on finite networks. A bottom layer and 20 layers above it each hold 20,000 firms, placed in 200 regions. Each firm picks its suppliers at random from the layer below, following its cluster pattern. We draw which regions are hit and work out which firms survive, layer by layer from the bottom. We compare each run with the theory after 20 steps of (2), evaluated at the share of regions actually hit in that run (with only 200 regions, this share varies). Over 3,360 runs (Figure 1), simulation and theory differ by 0.0025 on average. Gaps from the solid infinite-depth curves sit just past each threshold, where 20 layers are not enough for failures to spread all the way.

Sudden collapse (Q1). We ran 300 single-input networks ($n=4$, spread suppliers, $x=0.9$) at the ϕ^* predicted for each k (Figure 2a). With $k=1$, 90% of runs end with reliability below 0.1 and none above 0.21, so the network fades gradually. With $k=2$ the runs split into two groups: 20% collapse below 0.1, while 71% stay above 0.4.

Spread versus co-located (Q2). Table 1 estimates ϕ^* directly from the simulations, and every simulated value is within 0.004 of the 20-layer theory. With two inputs, co-location lowers the tolerable shock, as Claim 2 predicts. With one input and $k=2$, the extremes collapse together and the paired network collapses earlier than both, as in Example 1; Figure 2(c) shows a similar reversal for $n=6$, $k=3$. The 20-layer networks tolerate somewhat larger shocks than the infinite-depth theory, since a cascade has fewer layers to build up.

Firms versus supply (Q3). Making some suppliers far more popular than others (as in CNST's heavy-tailed data) changed average reliability by about 0.03 at most, which we expected, since in our model a popular supplier is no

Table 1: Tolerable shock ϕ^* ($n = 4$, $x = 0.9$). The simulated value is the realized shock that best separates collapsed runs (reliability below 0.1) from the rest, with a 95% bootstrap interval from 270 runs.

m	k	Geography c	Theory (infinite depth)	Theory (20 layers)	Simulation [95% CI]
2	1	4 (spread)	0.425	0.444	0.4425 [0.4400, 0.4500]
2	1	1 (co-located)	0.242	0.254	0.2575 [0.2525, 0.2600]
2	2	4 (spread)	0.077	0.086	0.0875 [0.0875, 0.0875]
2	2	2 (pairs)	0.047	0.054	0.0525 [0.0525, 0.0525]
2	2	1 (co-located)	0.039	0.045	0.0425 [0.0425, 0.0425]
1	2	4 (spread)	0.195	0.216	0.2175 [0.2175, 0.2175]
1	2	2 (pairs)	0.171	0.195	0.1925 [0.1925, 0.1975]
1	2	1 (co-located)	0.195	0.217	0.2175 [0.2175, 0.2200]

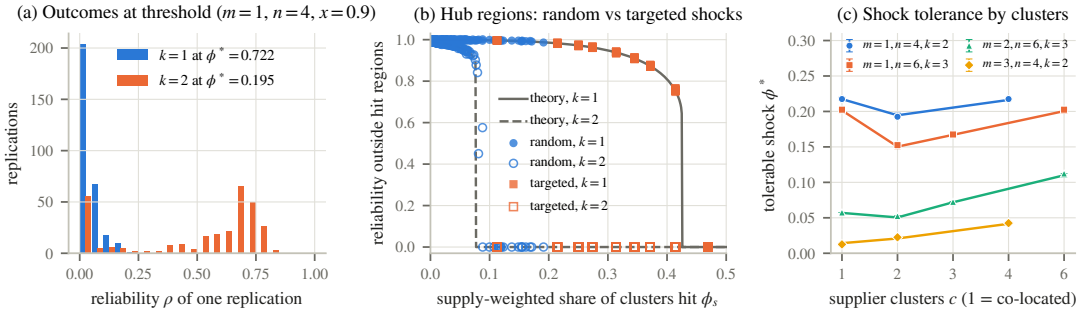


Figure 2: (a) Reliability in 300 runs at the theoretical threshold ($m = 1$): $k = 1$ fades, $k = 2$ splits into collapse and survival. (b) Supply concentrated in hub regions ($m = 2$, $n = 4$, $c = 4$, $x = 0.9$): runs follow the theory once the shock is measured by the share of supplier clusters hit. (c) Tolerable shock against the number of supplier clusters ($x = 0.9$): 20-layer theory (lines) and simulation with 95% intervals.

more likely to fail than any other. Concentrating supply in a few regions mattered much more. In one setup the largest region holds 11% of all supply relationships but only 0.5% of firms. Hitting that region alone collapses a $k = 2$ network, while hitting up to 10 random regions (5% of firms) collapses it in only 5% of runs. The runs line up with the theory (average error 0.002) if, instead of the share of regions hit, we plug in the share of all supplier clusters whose region was hit (Figure 2b). So the right measure of a regional shock is the share of supply it removes.

8 Further Research and Next Steps

Do firms choose fragile setups? (final project) In EGL, firms choose their relationship strength, and because they ignore how their reliability helps their customers, the equilibrium can sit right at the cliff. We want to let each firm also choose how many suppliers to keep and how many regions to place them in, paying a cost for each extra supplier and each extra region. In a symmetric equilibrium, each firm best-responds to the network's reliability, which is the solution of (2) at the chosen setup. We will compute best responses on a grid ($n \leq 6$, all splits c, a few cost levels) and compare the equilibrium with the setup a planner would choose. Our hypothesis is that firms under-diversify, and that when $k = 2$ some cost levels lead them to choose pairs, the worst split in Example 1.

Testing on data (final project). Our model suggests a better exposure measure for a CNST-style study: the share of each firm's suppliers *for each input* that sit in the disaster area, rather than the distance to it. The model predicts a sharp drop in outcomes once fewer than k suppliers survive, and more firm exits among the most exposed. CNST's data is proprietary, so we plan to start with public inter-country input–output tables (such as the OECD's), treating countries as regions. For a past regional disaster, such as the 2011 Thailand floods, we would compute each industry's share of input purchases from the hit region and check whether output losses rise sharply above some share rather than in proportion to it.

Smaller questions. We would like a simple condition on (n, k, m, x) that predicts when pairs are worst, results for networks with cycles, and a method for choosing which firms to protect under a budget. The last is a reverse influence-maximization problem [7] whose AND structure likely breaks submodularity.

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